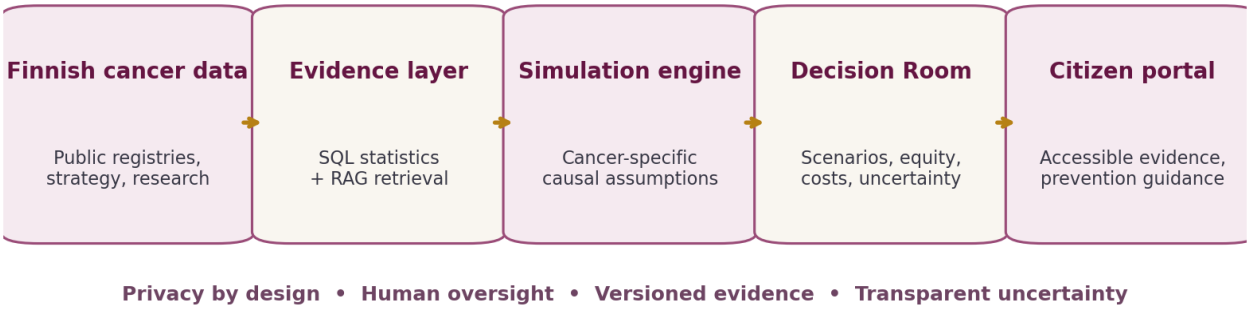
PRELIMINARY SUBMISSION | CONCEPT WHITE PAPER

Equal Care Navigator
The Decision Room

A National Cancer Equity & Policy Simulation Platform for Finland

“What if Finland could see the long-term consequences of healthcare decisions before making them?”

Simulate Tomorrow. Decide Better Today.



AstraZeneca Junction 2026 challenge | Independent team concept | Draft v0.1 | 13 November 2026

Visual direction: inspired by AstraZeneca’s publicly visible magenta, plum, warm gold and clean scientific editorial style. This is not an official AstraZeneca publication or an endorsement.

Executive Summary

Equal Care Navigator — The Decision Room is a proposed collaborative, AI-assisted policy simulation platform designed to help Finnish healthcare decision-makers evaluate the potential long-term consequences of cancer-prevention, screening and treatment decisions before committing public resources. Rather than merely reporting disparities, the system brings political leaders, wellbeing services county administrators, clinicians, public-health analysts and health economists into a shared, evidence-based decision process. Its central proposition is to turn population statistics, scientific evidence, uncertainty and policy trade-offs into a transparent, live scenario workshop.

A shared large-screen Decision Room enables participants to select policy scenarios, vary assumptions and compare plausible trajectories over five, ten and fifteen years. A citizen-facing experience explains approved public scenarios in accessible language without offering personal diagnoses or individual risk scores. The intended scope is all cancer types, with cancer-specific models introduced only when validated evidence is available.

The MVP will demonstrate a working session using real Finnish public aggregate screening and cancer statistics as baseline observations, one well-evidenced intervention, two or three counterfactual policy options, explicit uncertainty ranges and a source-grounded AI-generated decision brief. The application will never represent hypothetical figures as actual forecasts or treat correlation as proof of causation.

Privacy and public trust are foundational. The Junction MVP excludes individual patient records and relies on aggregated, legally reusable public data; subsequent use of any personal health information would trigger a separate legal, ethical, technical and governance assessment before collection or processing. Decisions remain with authorised humans.

1 The Policy Question & National Context

Finland's National Cancer Strategy 2026–2035 explicitly seeks to strengthen cancer prevention, earlier detection and equitable, effective cancer care. It recognises that cancer often develops over many years and that the health impact of preventive measures may become visible only in the longer term. This is the precise policy horizon that Equal Care Navigator intends to bring into the room: choices that look inexpensive or attractive today can carry delayed consequences for population outcomes, the distribution of benefits and the use of public funds. [1]

The Finnish health system is not federal. The relevant administrative landscape includes 21 wellbeing services counties, Helsinki and separate arrangements in Åland alongside national institutions. Regional and socioeconomic disparities are recognised policy concerns. However, differences in screening participation, diagnostic access or outcomes do not by themselves establish which intervention would cause an improvement. Our approach explicitly separates observed gaps, plausible mechanisms and modelled scenarios. [1–3]

2 The Proposed Answer to the AstraZeneca Challenge

Challenge requirement	Decision Room response
Identify the gap	Use published, comparable indicators to surface regional and socioeconomic disparities.
Locate the gap	Map evidence to prevention, screening, referral, diagnosis, treatment or follow-up.
Recommend what to do	Compare actionable, evidence-supported intervention scenarios and their delivery requirements.

Demonstrate impact	Show modelled ranges, assumptions, sensitivities and an evaluation KPI; validate prospectively.
Working solution	Deliver a live, interactive multi-user session rather than a slide-only concept.

The outcome is not an algorithm that allocates healthcare budgets autonomously. It is a structured decision environment that improves the quality, traceability and openness of human deliberation.

3 The Decision Room Experience

The Decision Room combines a shared interactive display or projector with individual participant devices. A facilitator opens a session; participants can inspect a common baseline, propose variations and review the evidence behind any modelled effect. Healthcare professionals challenge clinical assumptions, analysts examine the source data, public authorities assess deliverability and decision-makers balance health, equity and non-health public value.

A six-step facilitated workshop

- Frame the decision: define geography, cancer type(s), population, available budget and policy objective.
- Establish the baseline: display measured indicators with source, date, population denominator and limitations.
- Configure scenarios: change eligible programme funding, uptake, eligibility, capacity or outreach design.
- Simulate: compute the uncertainty-bounded evolution of health, economic and equity indicators.
- Deliberate: compare trade-offs, challenge assumptions and document dissent or missing evidence.
- Record the decision: produce an Action Card with rationale, owner, evaluation plan and human approval.

4 Example Policy Scenario: Competing Public Priorities

Consider an illustrative wellbeing services county discussion: a public capital investment such as a community library competes with funding for a preventive health programme. The library may have educational, civic and social benefits; prevention may have delayed health and economic returns. The Decision Room would not assume that one option is preferable. It would compare documented costs, plausible effects, distribution across income groups and consequences under alternative assumptions. Where the policy levers belong to different levels of government, the workshop must explicitly identify their separate legal budgets and decision rights.

A discussion about skin-cancer screening is useful to illustrate the interface but unsuitable as an assumed example of proven mortality reduction: evidence for population-wide visual skin examination among asymptomatic people is insufficient to establish a net benefit, and increased screening does not necessarily reduce cancer incidence. For the Junction demonstration, a more strongly evidenced screening or prevention intervention should be selected and assessed with Finnish experts. [6]

5 Simulation Methodology: No Forecast without Evidence

RAG retrieval is not a forecasting model. A reproducible simulation engine must combine actual baseline observations with intervention-effect estimates, demographic assumptions, epidemiological states and health-economic parameters. For suitable cancer types, a cohort Markov model, microsimulation or

another validated decision-analytic model could represent age groups, disease states, screening participation, detection, treatment and downstream costs. Specific model structure and effect parameters must be evidence-led, cancer-specific and independently reviewed.

A baseline describes a plausible “current policy continues” pathway. Counterfactual scenarios alter one or more clearly declared parameters. Parameter uncertainty should be represented through confidence or credible intervals, probabilistic sensitivity analysis and transparent uncertainty bands. Structural uncertainty, potential lead-time bias, overdiagnosis, unequal access, delayed effects and unmeasured confounding require explicit handling. Results should be presented as conditional projections — never deterministic prophecies or causal claims drawn merely from historical correlations.

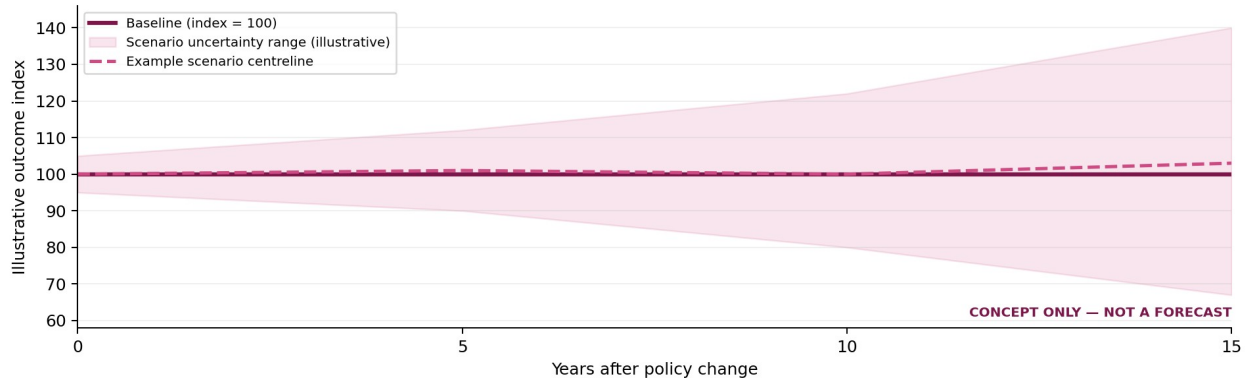


Figure: schematic comparison only. The plotted indices and uncertainty bands are synthetic explanatory artwork, not Finnish data or simulation outputs.

6 Data Foundation and Evidence Governance

The initial data catalogue includes the Finnish Cancer Registry, THL/Sotkanet, Statistics Finland, NORDCAN and the national strategy and implementation roadmap. Open published aggregates are held separately from narrative guidance and evidence. A structured statistical database stores measurements, definitions, years, geographic units, strata, numerators and denominators. A versioned retrieval index stores strategy passages, scientific references and intervention descriptions with their source pages. [1–5]

Ingestion gates verify public availability, legal reuse, provenance, population coverage, consistency and source version. Official strategic targets, actual observations, forecasts and preliminary releases must carry distinct data-status labels. If outcome data are absent for a cancer type or county, the system must report an evidence gap rather than infer missing values. Models require documented calibration and out-of-sample evaluation before their output is used to justify decisions.

7 Proposed Azure Technology Architecture

The technical foundation is an Azure-hosted React progressive web application with two role-specific views, served by Azure Static Web Apps, and a controlled Python/FastAPI backend on Azure Container Apps. Azure Database for PostgreSQL holds structured statistics and scenario metadata; Blob Storage preserves source files and approved input snapshots; Azure AI Search supports evidence retrieval; Azure OpenAI generates constrained explanations and draft Action Cards from deterministic model results and retrieved citations. Application Insights and Azure Monitor capture operational metrics while identity, secrets and access are secured by managed Azure services.

A versioned simulation service runs mathematical models independently of the language model. It must publish its inputs, outputs, uncertainty ranges, model version and limitations. Azure OpenAI is not authorised to invent impact estimates, access databases directly from the browser or recommend personal medical treatment. The backend enforces user roles, signed session actions, and limits on exported content. Deployment uses GitHub Actions with short-lived federated credentials and infrastructure as code. Region and service selection are subject to current EU availability and data-residency review.

8 GDPR-First Privacy and Security by Design

The strict protection of people and health data is a non-negotiable requirement. Health information is a special category of personal data under GDPR Article 9; a lawful basis under Article 6 alone is not sufficient for processing identifiable health information. For a public-data-only MVP, the team should avoid receiving or storing personal health records altogether. “Publicly accessible” does not automatically mean anonymous or legally reusable: datasets must be checked for re-identification risks, licence conditions and statistical disclosure rules. [7–9]

Minimum governance commitments

- Data minimisation and purpose limitation: no patient accounts or individual health histories in the initial analytical system.
- Public aggregate data only for the prototype, with small-cell suppression and controls against sensitive cross-tabulation or linkage.
- Explicit controller/processor roles, records of processing and vendor/transfer assessments before deployment in a real healthcare context.
- Encryption in transit and at rest, least-privilege role-based access, managed identities and restricted administrative exports.
- Separate development and production, secrets management, retention/deletion schedules and security testing.
- Human approval of public scenario outputs, no automated allocation of healthcare services or individual clinical advice.
- DPIA and security/privacy review where necessary before extending scope to personal data; test any intended AI use against EU AI Act and applicable medical-device rules.
- AI prompts, logs and analytics must not silently collect sensitive information; anonymisation and confidentiality claims must be technically verified.

The public portal should communicate only curated, source-linked, approved outputs and general preventive information. Any future personalised risk feature, patient-data integration or automated individual decision-making would require a new regulatory and clinical risk assessment, not merely a configuration change. “GDPR compliant” is not a certification that can be inferred from hosting in the EU. [7–10]

9 Junction 2026 Minimum Viable Demonstration

The preliminary submission proposes a focused working demonstration that makes the broad national vision credible without overstating what can be built or validated during a hackathon. The platform is designed for all cancers at the data and navigation level, but an individual simulation is only activated where an appropriate scientific model and defensible intervention-effect parameters exist.

- A shared Decision Room with real-time scenario selection and a large-screen, accessible dashboard.

- Real Finnish public cancer and screening statistics as the source-labelled baseline, imported from the team's verified repository datasets.
- One well-studied cancer prevention or screening intervention with documented clinical effect and model assumptions.
- Two or three alternative public-policy scenarios, including explicit uncertainty and distributional effects across population groups.
- An Azure OpenAI-produced decision brief that cites its sources, distinguishes measured from simulated results and explains model limitations.

Success is defined by a repeatable end-to-end demonstration: choose a baseline, change a policy lever, run the validated calculation, explain uncertainty, compare equity implications and generate a human-reviewable Action Card. The team must be able to show the provenance of every displayed real statistic and model parameter.

10 Impact, Evaluation and Responsible Governance

The central value proposition is to consider preventive investment, long-term healthcare costs and social justice together. Economic evaluation can incorporate public-sector budget impact, cost-effectiveness where appropriate and non-health opportunity costs, without pretending that social benefits and clinical outcomes are directly interchangeable. Decision-makers should see who benefits, who may be disadvantaged, which assumptions dominate the result and where additional evidence has the highest value.

Evaluation will monitor predictive calibration against subsequent published observations, stability across regions, stakeholder comprehension, evidence coverage, model documentation, subgroup fairness and reproducibility. The workshop outcome is a documented deliberation, not an automatic budget recommendation. The design explicitly allows the "best current decision" to be "collect better evidence first".

11 Delivery Roadmap & Critical Risks

Stage	Output	Quality gate
Foundation	Verified public statistics and source register	Provenance and licences checked
Simulation core	Cancer-specific baseline and intervention model	Expert-reviewed assumptions and uncertainty
Decision Room MVP	Collaborative PWA, scenario controls and graphs	End-to-end reproducible workshop
AI explanation	Source-grounded Action Card	No fabricated numbers, clear caveats
Governance & pilot	Access controls, logs, privacy plan	DPIA decision, security review, clinical oversight
Scale-up	Additional cancers, counties, policy levers	Per-model validation and stakeholder approval

Major risks include inconsistent open data, insufficient Finnish effect-size evidence, falsely precise projections, misinterpretation by politicians or citizens, unauthorised sharing of sensitive meeting content, and potential conflicts between health and other public objectives. Controls include source traceability, scenario watermarking, model validation, sensitivity analysis, privacy reviews, role-based sharing and explicit human ownership of decisions.

12 Conclusion

Equal Care Navigator — The Decision Room transforms the policy question “What should Finland do?” into a structured, collaborative investigation of “What might happen under each plausible choice, to whom, at what cost and with what uncertainty?” It makes the long horizon of cancer prevention visible in the present, while respecting the limits of evidence and the authority of democratic and medical decision-makers.

The next step is not a promise of perfectly predicting cancer outcomes. It is a credible demonstration of an evidence-grounded decision process: source-linked Finnish statistics, one defensible intervention model, alternative scenarios, honest uncertainty, GDPR-first governance and a shared room in which decision-makers can examine the future before choosing a course of action.

References and Evidence Notes

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